

Paper Code:BCA 106				Paper: Data Structure and Algorithms Using C				L	T/P	C		
								4	-	4		
Marking Scheme: 1. Teachers Continuous Evaluation: 40 marks 2. Term end Theory Examinations: 60 marks												
Instruction for paper setter: 1. There should be 9 questions in the term end examinations question paper. 2. The first (1 st) question should be compulsory and cover the entire syllabus. This question should be objective, single line answers or short answer type question of total 12 marks. 3. Apart from question 1 which is compulsory, rest of the paper shall consist of 4 units as per the syllabus. Every unit shall have two questions covering the corresponding unit of the syllabus. However, the student shall be asked to attempt only one of the two questions in the unit. Individual questions may contain upto 5 sub-parts / sub-questions. Each Unit shall have a marks weightage of 12. 4. The questions are to be framed keeping in view the learning outcomes of the course / paper. The standard / level of the questions to be asked should be at the level of the prescribed textbook. 5. The requirement of (scientific) calculators / log-tables / data – tables may be specified if required.												
Course Outcomes (CO):												
CO1		Familiarize the basics of data structures and algorithms.										
CO2		Understand and apply linear and nonlinear data structures and their operations.										
CO3		Compare and implement searching, sorting and hashing techniques.										
CO4		Appraise and determine the correct data structure for any given real world problem.										
Course Outcomes (CO to Programme Outcomes (PO) Mapping (scale 1: low, 2: Medium, 3: High												
CO/PO	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12
CO1	3	–	2	–	–	–	–	–	–	–	1	2
CO2	3	–	3	2	–	–	–	–	–	–	–	2
CO3	2	–	3	2	–	–	–	–	–	–	–	2
CO4	2	–	3	3	–	–	–	2	–	–	1	2

UNIT – I

No. of Hours: 11

Linear Data Structures- Static: Introduction to Algorithms- Attributes, Design Techniques, Time Space Trade Off, Data Structures, Classification and Operations of Data Structures. Growth of functions, Masters Theorem

Arrays: Single Dimension, Two-Dimension and Introduction to Multi Dimensions, Memory Representation, Address Calculation, Sparse Matrices- Types, Representation.

Introduction: Dynamic Memory Allocation, Dynamic Memory versus Static Memory Allocation.

Linked List Types: Singly Linked List, Doubly Linked List, Header Linked List, Circular Linked List.

Operations: Creation, Insertion, Deletion, Modification, Searching, Sorting, Reversing, and Merging.

UNIT – II

No. of Hours: 11

Searching and Sorting: Linear and Binary Search, Selection Sort, Bubble Sort, Insertion Sort, Merge Sort, quicksort, Elementary Comparison of Searching and Sorting Algorithms.

Stacks: Introduction, Static and Dynamic Implementation, Operations, Applications- Evaluation and Conversion between Polish and Reverse Polish Notations.

Queues: Introduction, Static and Dynamic Implementation, Operations, Types- Linear Queue, Circular Queue, Doubly Ended Queue, Priority Queue.

Hashing: Hash Table, Hash Functions, and Collision Resolution by chaining and double hashing.

UNIT – III

No. of Hours: 10

Trees: Notations & Terminologies, Memory Representation, Binary Trees Types- Complete, Full, Strict, Expression Binary Tree, Evaluation of arithmetic expressions using expression binary trees. Tree Traversals (Recursive and non-recursive) , Binary Search Tree, AVL Tree. M-Way trees, B-Trees.

UNIT – IV

No. of Hours: 10

Introduction to Graphs: Notations & Terminologies, Representation of Graphs- Adjacency Matrix, Incidence Matrix and Linked Representation. Graph Traversal (Breadth First and Depth First), Prim's Algorithm, Kruskal's algorithm, Dijkstra's Algorithm, Bellman-Ford algorithm, Warshall's Algorithm.

TEXT BOOKS:

TB1. Y. Langsam, M. J. Augenstein and A.M. Tanenebaum, "Data Structures using C and C++", Pearson Education India, Second Edition, 2015.

TB2. E. Horowitz and S. Sahni, "Fundamentals of Data Structures in C". Universities Press, Second edition, 2008.

REFERENCE BOOKS:

RB1. Schaum's Outline Series, "Data Structures", TMH, Special Indian Ed., Seventeenth Reprint, 2014.

RB2. Ashok N kamthane "Introduction to Data Structures in C", Pearson, Third Edition, 2009.

RB3. D. Samanta, "Classic Data Structures", PHI, Second Edition, 2009.

RB4. D. Malhotra and N. Malhotra, "Data Structures and Program Design using C", Laxmi Publications, Indian adapted edition from Mercury Learning and Information-USA, First edition, 2018.

RB5. Y. Kanetkar "Data Structures through C", BPB Publication, Third Edition, 2019.

RB6. R.F Gilberg, and B A Frouzan- "Data Structures: A Pseudocode Approach with C", Thomson Learning, Second Edition, 2004.

RB7. A. K. Rath, and A.K. Jagadev, "Data Structures and Program Design Using C", Scitech Publications, Second Edition, 2011.